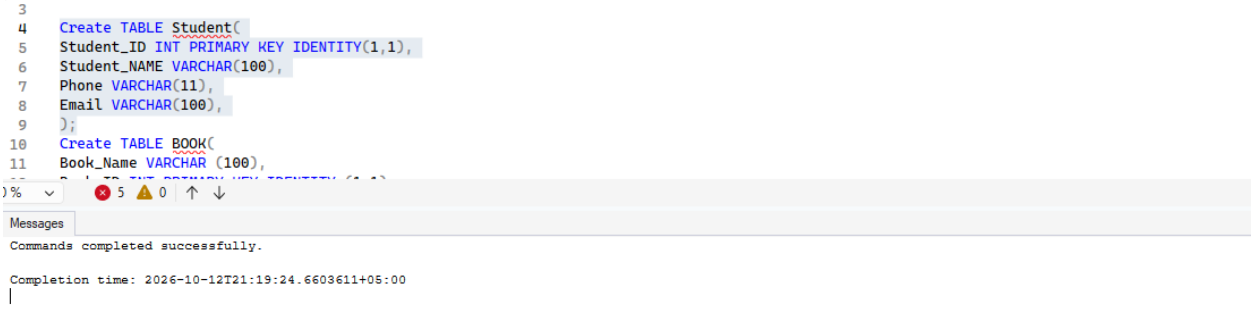
NAME:Umair Iftikhar
SAPID:63825
SUBJECT:DATA WAREHOUSE

# LAB TASK#5

## 1) DDL — Data Definition Language Query

1) CREATE — Create a table

```
Create TABLE Student(
Student_ID INT PRIMARY KEY IDENTITY(1,1),
Student_NAME VARCHAR(100),
Phone VARCHAR(11),
Email VARCHAR(100),
);
```




2) ALTER — Add a column to an existing table

```
ALTER TABLE Book
ADD Author_Name VARCHAR(100);
```

NAME:Umair Iftikhar
SAPID:63825
SUBJECT:DATA WAREHOUSE

```
135 - DESKTOP-
90    ALTER TABLE Book
91    ADD Author_Name VARCHAR(100);
92    SELECT * FROM Book;
```

100 % 5 0

Results Messages

| | Book_Name | Book_ID | Author_Name |
|---|---|---|---|
| 1 | Computer Learning | 1 | NULL |
| 2 | Football Technique | 2 | NULL |
| 3 | SQL Query | 3 | NULL |
| 4 | Computer Learning | 4 | NULL |
| 5 | Football Technique | 5 | NULL |
| 6 | SQL Query | 6 | NULL |

3) TRUNCATE — Remove all records from a table

```
TRUNCATE TABLE Fine;
```

```
92    SELECT * FROM Book;
93    TRUNCATE TABLE Fine;
94    select * From Fine;
```

100 % 5 0

Results Messages

| Fine_ID | Student_ID | Book_ID | Amount | Fine_Date | Status |
|---|---|---|---|---|---|

4) DROP — Delete a table

```
DROP TABLE Fine;
```

NAME:Umair Iftikhar
SAPID:63825
SUBJECT:DATA WAREHOUSE

```
97    DROP TABLE Fine;
98    select * From Fine;
```

100 % 6 0

Messages

```
Msg 208, Level 16, State 1, Line 98
Invalid object name 'Fine'.

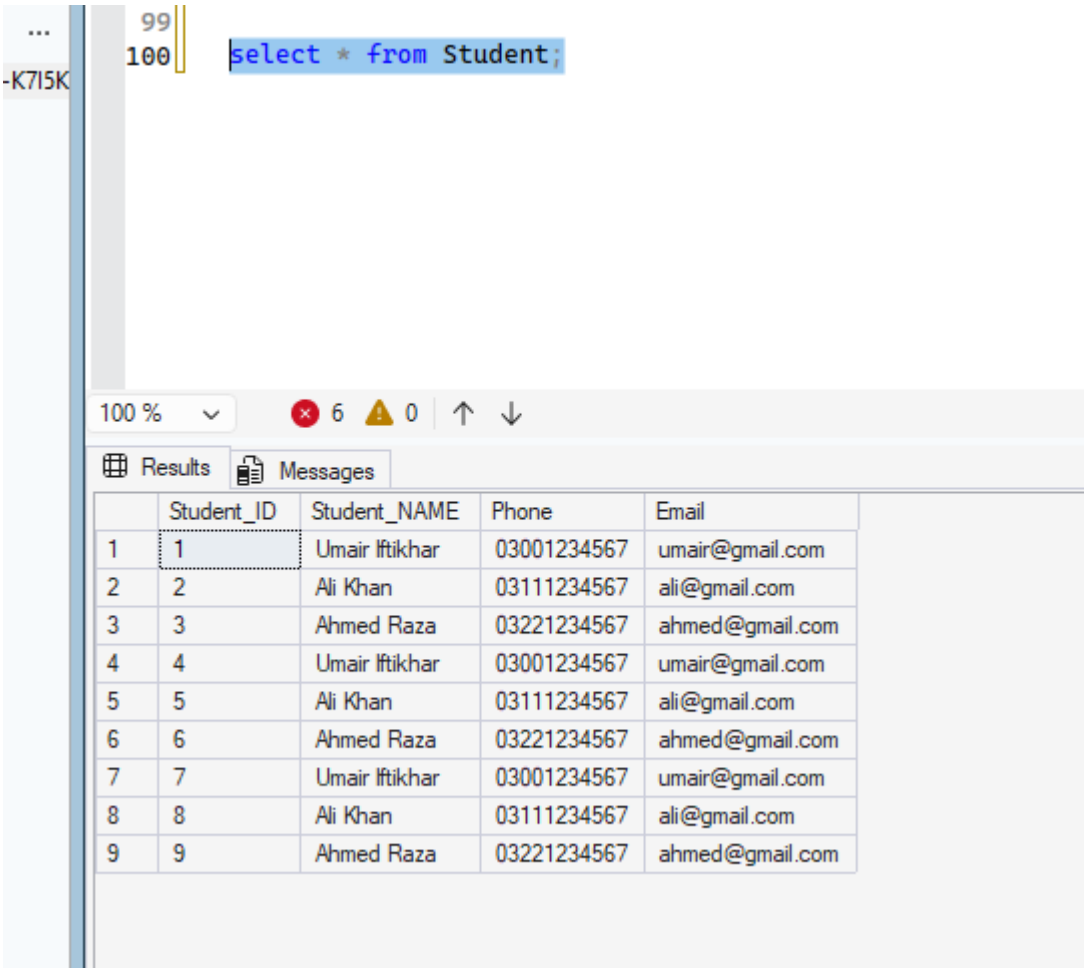
Completion time: 2026-10-12T21:37:57.9594940+05:00
```

## 2) DML — Data Manipulation Language

1)INSERT — Add a student

```
INSERT INTO Student
(Student_Name, Email, Phone )
VALUES
('Umair Iftikhar', 'umair@gmail.com', '03001234567' ),
('Ali Khan', 'ali@gmail.com', '03111234567' ),
('Ahmed Raza', 'ahmed@gmail.com', '03221234567');
```

NAME:Umair Iftikhar
SAPID:63825
SUBJECT:DATA WAREHOUSE




| | Student_ID | Student_NAME | Phone | Email |
|---|---|---|---|---|
| 1 | 1 | Umair Iftikhar | 03001234567 | umair@gmail.com |
| 2 | 2 | Ali Khan | 03111234567 | ali@gmail.com |
| 3 | 3 | Ahmed Raza | 03221234567 | ahmed@gmail.com |
| 4 | 4 | Umair Iftikhar | 03001234567 | umair@gmail.com |
| 5 | 5 | Ali Khan | 03111234567 | ali@gmail.com |
| 6 | 6 | Ahmed Raza | 03221234567 | ahmed@gmail.com |
| 7 | 7 | Umair Iftikhar | 03001234567 | umair@gmail.com |
| 8 | 8 | Ali Khan | 03111234567 | ali@gmail.com |
| 9 | 9 | Ahmed Raza | 03221234567 | ahmed@gmail.com |

2) UPDATE — Change a student's phone number

```
UPDATE Student
SET Phone = '03123456789'
WHERE Student_ID = 1;
```

NAME:Umair Iftikhar
SAPID:63825
SUBJECT:DATA WAREHOUSE

.1135 - DESKTOP-K7I5K

```
101
102   UPDATE Student
103   SET Phone = '03123456789'
104   WHERE Student_ID = 1;
105   select * from Student;
```

100 % 6 0

Results Messages

| | Student_ID | Student_NAME | Phone | Email |
|---|---|---|---|---|
| 1 | 1 | Umair Iftikhar | 03123456789 | umair@gmail.com |
| 2 | 2 | Ali Khan | 03111234567 | ali@gmail.com |
| 3 | 3 | Ahmed Raza | 03221234567 | ahmed@gmail.com |
| 4 | 4 | Umair Iftikhar | 03001234567 | umair@gmail.com |
| 5 | 5 | Ali Khan | 03111234567 | ali@gmail.com |
| 6 | 6 | Ahmed Raza | 03221234567 | ahmed@gmail.com |
| 7 | 7 | Umair Iftikhar | 03001234567 | umair@gmail.com |
| 8 | 8 | Ali Khan | 03111234567 | ali@gmail.com |
| 9 | 9 | Ahmed Raza | 03221234567 | ahmed@gmail.com |

3) DELETE — Delete a student

```
DELETE FROM Student
WHERE Student_ID = 3;
```

NAME:Umair Iftikhar
SAPID:63825
SUBJECT:DATA WAREHOUSE

```
107    DELETE FROM Student
108    WHERE Student_ID = 3;
109    select * from Student;
```

100 % 6 0

Results Messages

| | Student_ID | Student_NAME | Phone | Email |
|---|---|---|---|---|
| 1 | 1 | Umair Iftikhar | 03123456789 | umair@gmail.com |
| 2 | 2 | Ali Khan | 03111234567 | ali@gmail.com |
| 3 | 4 | Umair Iftikhar | 03001234567 | umair@gmail.com |
| 4 | 5 | Ali Khan | 03111234567 | ali@gmail.com |
| 5 | 6 | Ahmed Raza | 03221234567 | ahmed@gmail.com |
| 6 | 7 | Umair Iftikhar | 03001234567 | umair@gmail.com |
| 7 | 8 | Ali Khan | 03111234567 | ali@gmail.com |
| 8 | 9 | Ahmed Raza | 03221234567 | ahmed@gmail.com |

4) SELECT — Display student records

```
select * from Student;
```

```
100    select * from Student;
```

100 % 6 0

Results Messages

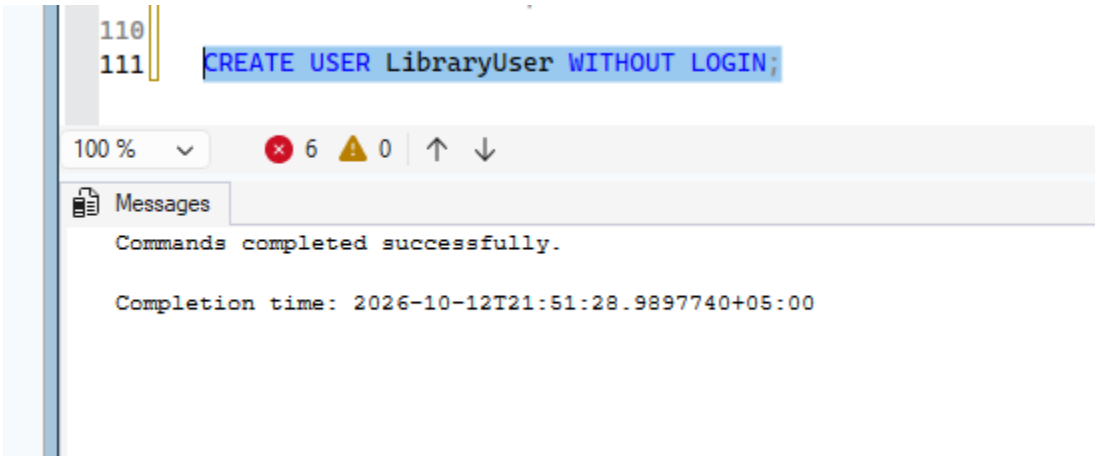
| | Student_ID | Student_NAME | Phone | Email |
|---|---|---|---|---|
| 1 | 1 | Umair Iftikhar | 03123456789 | umair@gmail.com |
| 2 | 2 | Ali Khan | 03111234567 | ali@gmail.com |
| 3 | 3 | Ahmed Raza | 03221234567 | ahmed@gmail.com |
| 4 | 4 | Umair Iftikhar | 03001234567 | umair@gmail.com |
| 5 | 5 | Ali Khan | 03111234567 | ali@gmail.com |
| 6 | 6 | Ahmed Raza | 03221234567 | ahmed@gmail.com |
| 7 | 7 | Umair Iftikhar | 03001234567 | umair@gmail.com |
| 8 | 8 | Ali Khan | 03111234567 | ali@gmail.com |
| 9 | 9 | Ahmed Raza | 03221234567 | ahmed@gmail.com |

## 3) DCL — Data Control Language

1)CREATE USER — Create a database user

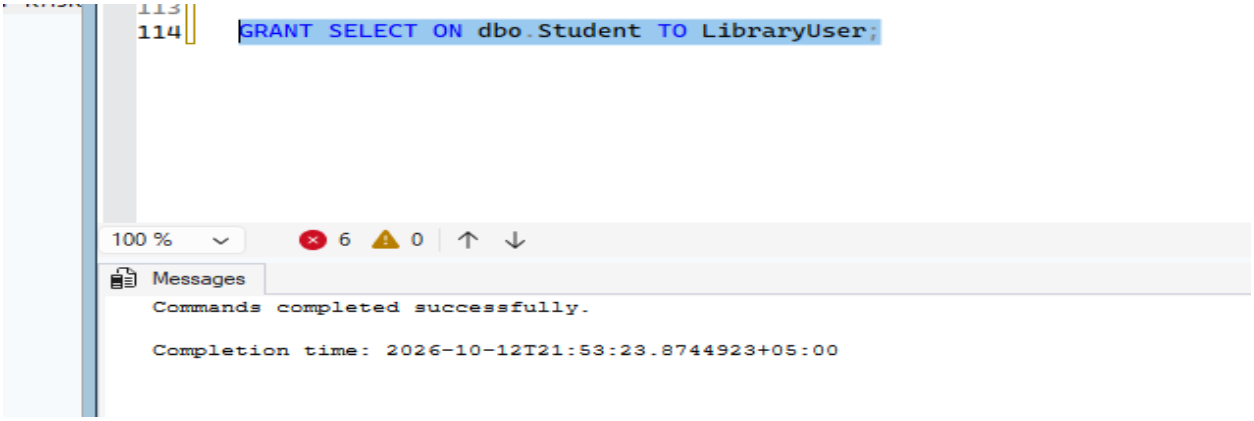
```
CREATE USER LibraryUser WITHOUT LOGIN;
```

NAME:Umair Iftikhar
SAPID:63825
SUBJECT:DATA WAREHOUSE



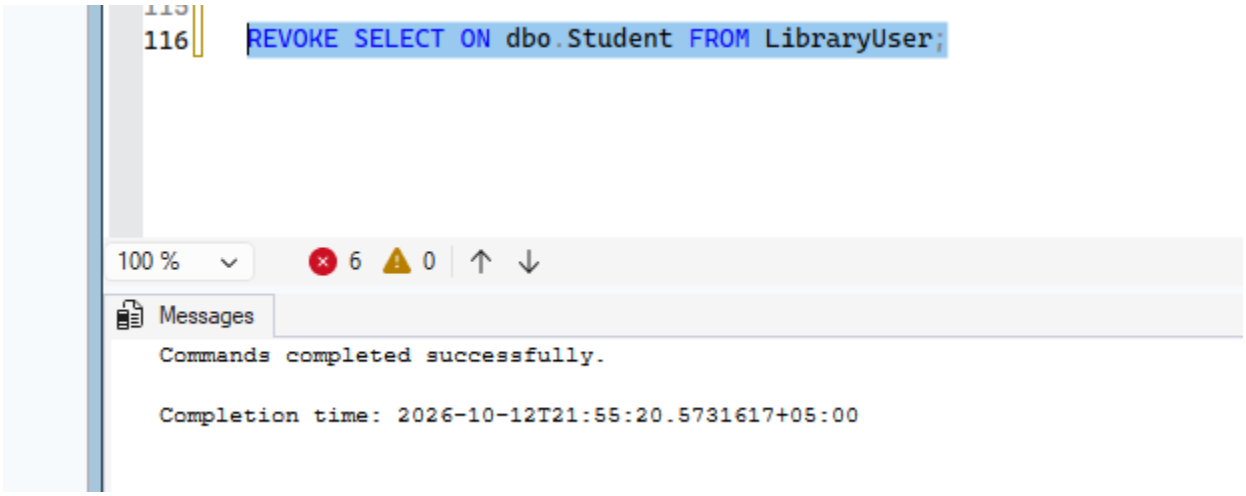

2) GRANT — Give permission to view student records

GRANT SELECT ON dbo.Student TO LibraryUser;




3) REVOKE — Remove the permission

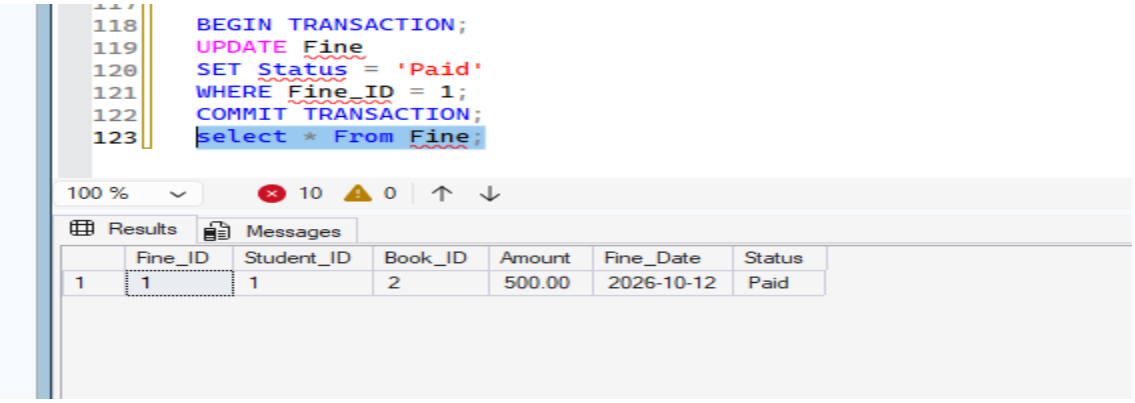
REVOKE SELECT ON dbo.Student FROM LibraryUser;




## 4) TCL — Transaction Control Language

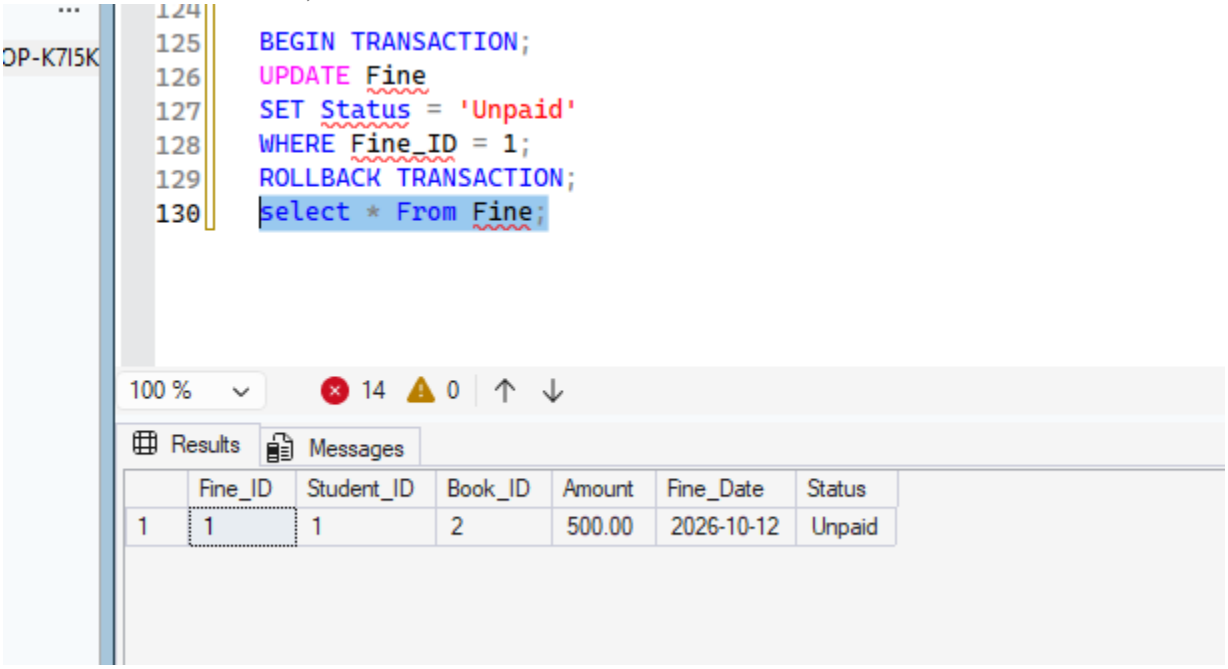
NAME:Umair Iftikhar
SAPID:63825
SUBJECT:DATA WAREHOUSE
1)COMMIT — Save a change permanently

```
BEGIN TRANSACTION;
UPDATE Fine
SET Status = 'Paid'
WHERE Fine_ID = 1;
COMMIT TRANSACTION;
```




2) ROLLBACK — Undo a transaction

```
BEGIN TRANSACTION;
UPDATE Fine
SET Status = 'Unpaid'
WHERE Fine_ID = 1;
ROLLBACK TRANSACTION;
```

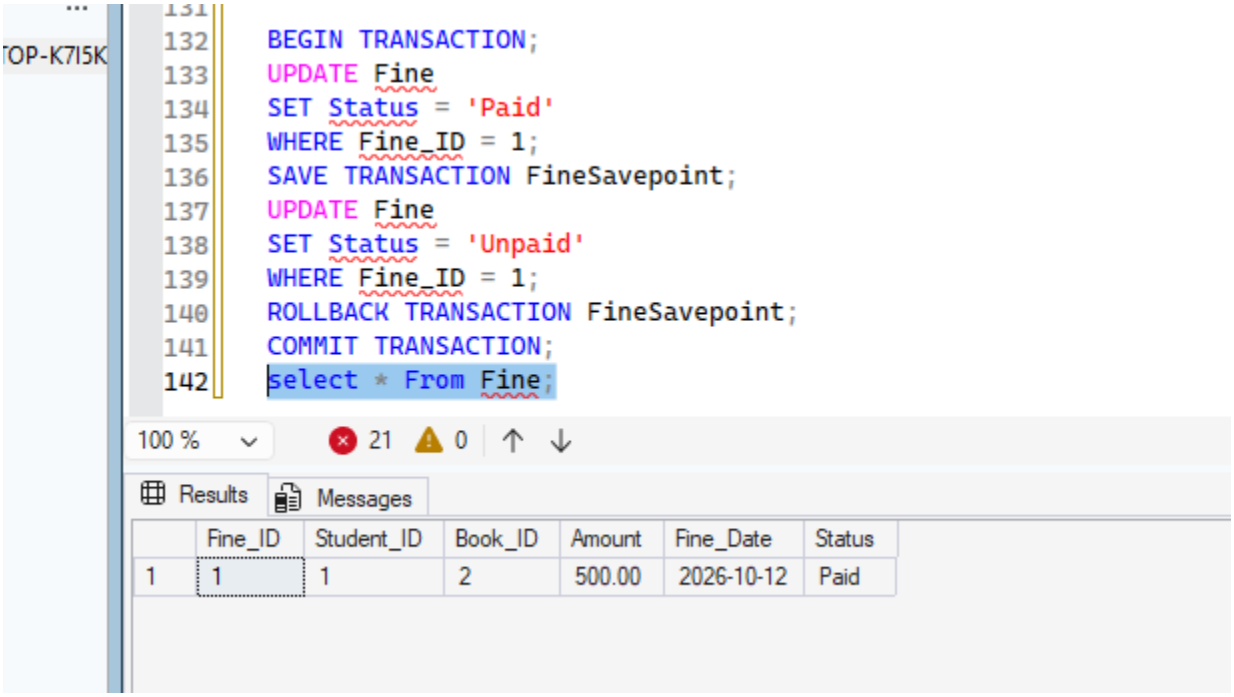



3) SAVE

```
BEGIN TRANSACTION;
```

NAME:Umair Iftikhar
SAPID:63825
SUBJECT:DATA WAREHOUSE

```
UPDATE Fine
SET Status = 'Paid'
WHERE Fine_ID = 1;
SAVE TRANSACTION FineSavepoint;
UPDATE Fine
SET Status = 'Unpaid'
WHERE Fine_ID = 1;
ROLLBACK TRANSACTION FineSavepoint;
COMMIT TRANSACTION;
```

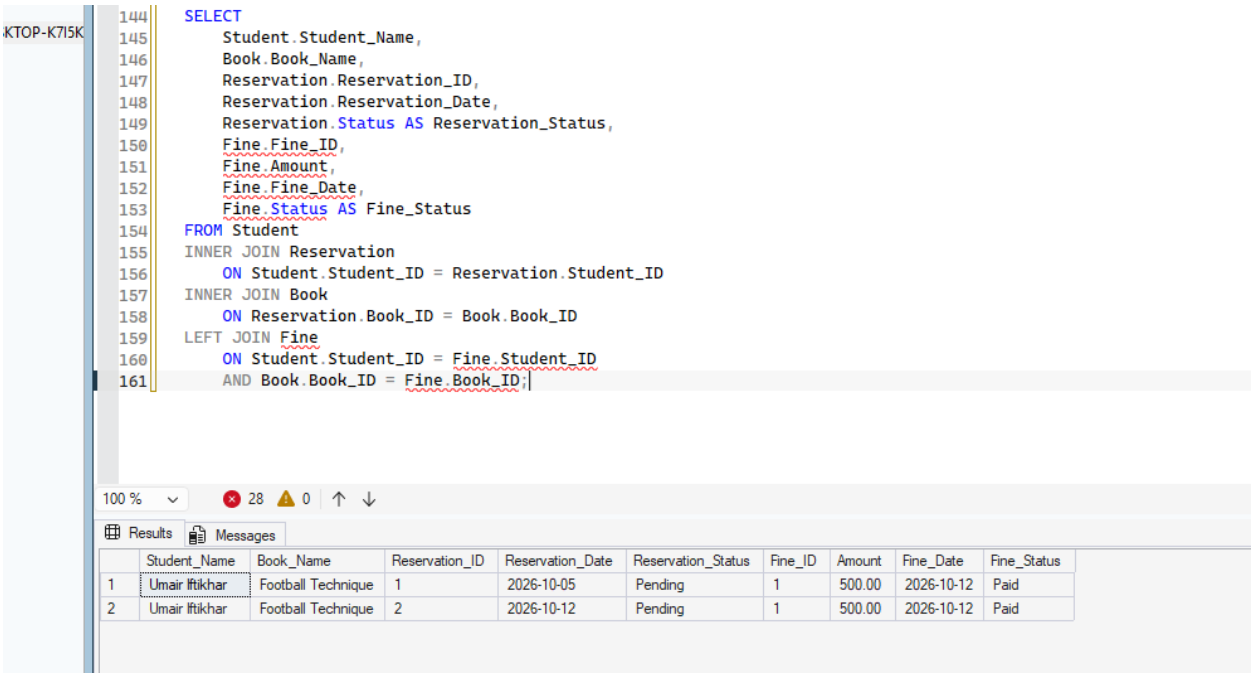



## 5) JOIN — Reservation details and fine details

```
SELECT
    Student.Student_Name,
    Book.Book_Name,
    Reservation.Reservation_ID,
    Reservation.Reservation_Date,
    Reservation.Status AS Reservation_Status,
    Fine.Fine_ID,
    Fine.Amount,
    Fine.Fine_Date,
    Fine.Status AS Fine_Status
FROM Student
INNER JOIN Reservation
    ON Student.Student_ID = Reservation.Student_ID
```

NAME:Umair Iftikhar
SAPID:63825
SUBJECT:DATA WAREHOUSE

```
INNER JOIN Book
    ON Reservation.Book_ID = Book.Book_ID
LEFT JOIN Fine
    ON Student.Student_ID = Fine.Student_ID
    AND Book.Book_ID = Fine.Book_ID;
```




## 6) GROUP BY — Reservations and fines per student

```
SELECT
    Student.Student_Name,
    Book.Book_Name,
    Reservation.Reservation_ID,
    Reservation.Reservation_Date,
    Reservation.Status AS Reservation_Status,
    Fine.Fine_ID,
    Fine.Amount,
    Fine.Fine_Date,
    Fine.Status AS Fine_Status
FROM Student
INNER JOIN Reservation
    ON Student.Student_ID = Reservation.Student_ID
INNER JOIN Book
    ON Reservation.Book_ID = Book.Book_ID
LEFT JOIN Fine
    ON Student.Student_ID = Fine.Student_ID
    AND Book.Book_ID = Fine.Book_ID;
    SELECT
    Student.Student_ID,
    Student.Student_Name,
```

NAME:Umair Iftikhar
SAPID:63825
SUBJECT:DATA WAREHOUSE

```
    COUNT(DISTINCT Reservation.Reservation_ID) AS Total_Reservations,
    COUNT(DISTINCT Fine.Fine_ID) AS Total_Fines,
    ISNULL(SUM(DISTINCT Fine.Amount), 0) AS Total_Fine_Amount
FROM Student
LEFT JOIN Reservation
    ON Student.Student_ID = Reservation.Student_ID
LEFT JOIN Fine
    ON Student.Student_ID = Fine.Student_ID
GROUP BY
    Student.Student_ID,
    Student.Student_Name;
```

```
SELECT
    Student.Student_Name,
    Book.Book_Name,
    Reservation.Reservation_ID,
    Reservation.Reservation_Date,
    Reservation.Status AS Reservation_Status,
    Fine.Fine_ID,
    Fine.Amount,
    Fine.Fine_Date,
    Fine.Status AS Fine_Status
FROM Student
INNER JOIN Reservation
    ON Student.Student_ID = Reservation.Student_ID
INNER JOIN Book
    ON Reservation.Book_ID = Book.Book_ID
LEFT JOIN Fine
    ON Student.Student_ID = Fine.Student_ID
    AND Book.Book_ID = Fine.Book_ID;
    SELECT
    Student.Student_ID,
```

100 % 32 0

Results Messages

| | Student_Name | Book_Name | Reservation_ID | Reservation_Date | Reservation_Status | Fine_ID | Amount | Fine_Date | Fine_Status |
|---|---|---|---|---|---|---|---|---|---|
| 1 | Umair Iftikhar | Football Technique | 1 | 2026-10-05 | Pending | 1 | 500.00 | 2026-10-12 | Paid |
| 2 | Umair Iftikhar | Football Technique | 2 | 2026-10-12 | Pending | 1 | 500.00 | 2026-10-12 | Paid |

| | Student_ID | Student_Name | Total_Reservations | Total_Fines | Total_Fine_Amount |
|---|---|---|---|---|---|
| 1 | 1 | Umair Iftikhar | 2 | 1 | 500.00 |
| 2 | 2 | Ali Khan | 0 | 0 | 0.00 |
| 3 | 4 | Umair Iftikhar | 0 | 0 | 0.00 |
| 4 | 5 | Ali Khan | 0 | 0 | 0.00 |
| 5 | 6 | Ahmed Raza | 0 | 0 | 0.00 |
| 6 | 7 | Umair Iftikhar | 0 | 0 | 0.00 |
| 7 | 8 | Ali Khan | 0 | 0 | 0.00 |
| 8 | 9 | Ahmed Raza | 0 | 0 | 0.00 |

## 7) HAVING — Students with more than one reservation

```
   SELECT
   Student.Student_ID,
   Student.Student_Name,
 COUNT(Reservation.Reservation_ID) AS Total_Reservations
FROM Student
INNER JOIN Reservation
    ON Student.Student_ID = Reservation.Student_ID
GROUP BY
    Student.Student_ID,
    Student.Student_Name
HAVING COUNT(Reservation.Reservation_ID) > 1;
```

NAME:Umair Iftikhar
SAPID:63825
SUBJECT:DATA WAREHOUSE

DESKTOP-K7I5K

```
177
178         SELECT
179         Student.Student_ID,
180         Student.Student_Name,
181         COUNT(Reservation.Reservation_ID) AS Total_Reservations
182     FROM Student
183     INNER JOIN Reservation
184         ON Student.Student_ID = Reservation.Student_ID
185     GROUP BY
186         Student.Student_ID,
187         Student.Student_Name
188     HAVING COUNT(Reservation.Reservation_ID) > 1;
```

100 % 32 0

Results Messages

| | Student_ID | Student_Name | Total_Reservations |
|---|---|---|---|
| 1 | 1 | Umair Iftikhar | 2 |